

## WHITE GRASS TANNA ISLAND, Vanuatu

WMO: 915650

Lat: 19.455S

Lon: 169.224E

Elev: 6

StdP: 101.26

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 8             | 16.0       | 16.7 | 12.3                          | 8.9 | 18.7 | 13.3 | 9.5 | 19.2 | 9.4                   | 24.3 | 9.0 | 24.3 | 0.7                   | 120  | 0.270 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 6.6                    | 31.4            | 26.7 | 30.9 | 26.5 | 30.3 | 26.1 | 28.0                | 30.4 | 27.4 | 30.0 | 26.9 | 29.6 | 4.0                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 27.2                            | 23.0 | 30.0 | 26.7 | 22.3 | 29.6 | 26.1 | 21.5 | 29.0 | 90.1          | 30.7 | 87.3 | 30.1 | 84.9 | 29.6 | 30.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 8.9           | 8.1  | 7.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 24.4   | 26.4  | 27.0  | 26.7  | 25.6  | 24.0  | 22.7  | 21.9  | 21.8  | 22.6  | 23.7  | 24.7  | 25.9 |      |
|                                                                               | DBStd                                                                         | 2.28   | 1.08  | 1.04  | 0.97  | 1.33  | 1.46  | 1.71  | 1.49  | 1.71  | 1.53  | 1.48  | 1.40  | 1.28 |      |
|                                                                               | HDD10.0                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                               | HDD18.3                                                                       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                               | CDD10.0                                                                       | 5252   | 507   | 476   | 517   | 468   | 433   | 381   | 371   | 365   | 378   | 424   | 440   | 493  |      |
|                                                                               | CDD18.3                                                                       | 2211   | 249   | 242   | 258   | 218   | 175   | 131   | 112   | 107   | 128   | 166   | 190   | 235  |      |
|                                                                               | CDH23.3                                                                       | 16599  | 2354  | 2418  | 2474  | 1754  | 1004  | 568   | 371   | 383   | 644   | 1083  | 1428  | 2117 |      |
|                                                                               | CDH26.7                                                                       | 3801   | 658   | 761   | 739   | 392   | 115   | 42    | 20    | 18    | 62    | 158   | 267   | 569  |      |
| Wind                                                                          | WSAvg                                                                         | 2.6    | 2.7   | 2.5   | 2.2   | 2.3   | 2.3   | 2.5   | 2.5   | 2.9   | 2.9   | 3.0   | 2.8   | 2.8  |      |
| Precipitation                                                                 | PrecAvg                                                                       | 2026   | 263   | 220   | 278   | 269   | 193   | 98    | 106   | 108   | 96    | 114   | 124   | 170  |      |
|                                                                               | PrecMax                                                                       | 2933   | 588   | 573   | 608   | 555   | 415   | 245   | 231   | 257   | 310   | 325   | 319   | 331  |      |
|                                                                               | PrecMin                                                                       | 1409   | 93    | 59    | 120   | 141   | 92    | 31    | 46    | 58    | 32    | 48    | 56    | 84   |      |
|                                                                               | PrecStd                                                                       | 370    | 130   | 119   | 120   | 118   | 87    | 60    | 48    | 52    | 73    | 61    | 63    | 66   |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 32.1  | 32.2  | 32.2  | 31.2  | 29.4  | 28.9  | 28.3  | 28.2  | 29.4  | 30.2  | 30.6  | 31.2 |      |
|                                                                               |                                                                               | MCWB   | 26.6  | 27.6  | 27.2  | 26.7  | 25.6  | 24.7  | 24.3  | 23.4  | 23.6  | 24.7  | 25.3  | 25.8 |      |
|                                                                               | 2%                                                                            | DB     | 31.0  | 31.4  | 31.1  | 30.2  | 28.8  | 27.9  | 27.2  | 27.0  | 28.2  | 29.1  | 29.9  | 30.6 |      |
|                                                                               |                                                                               | MCWB   | 26.0  | 27.2  | 27.1  | 26.4  | 25.2  | 23.8  | 23.2  | 22.5  | 23.3  | 24.0  | 24.9  | 25.6 |      |
|                                                                               | 5%                                                                            | DB     | 30.2  | 30.9  | 30.5  | 29.7  | 28.0  | 27.0  | 26.2  | 26.1  | 27.3  | 28.2  | 29.1  | 30.1 |      |
|                                                                               |                                                                               | MCWB   | 25.6  | 26.9  | 26.8  | 26.0  | 24.5  | 23.2  | 22.5  | 22.0  | 22.7  | 23.5  | 24.3  | 25.5 |      |
|                                                                               | 10%                                                                           | DB     | 29.7  | 30.2  | 29.9  | 28.9  | 27.2  | 26.2  | 25.3  | 25.4  | 26.6  | 27.5  | 28.4  | 29.5 |      |
|                                                                               |                                                                               | MCWB   | 25.4  | 26.4  | 26.5  | 25.6  | 23.8  | 22.7  | 21.7  | 21.5  | 22.2  | 23.0  | 24.0  | 25.1 |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 27.7  | 29.3  | 28.9  | 27.8  | 26.7  | 25.5  | 25.4  | 24.9  | 25.1  | 25.9  | 26.8 | 27.2 |
|                                                                               |                                                                               |        | MCDB  | 30.8  | 31.6  | 30.9  | 30.0  | 28.7  | 27.9  | 27.4  | 27.1  | 27.9  | 28.9  | 29.4 | 29.9 |
| 2%                                                                            |                                                                               | WB     | 27.0  | 28.1  | 28.1  | 27.1  | 25.9  | 24.7  | 24.1  | 23.7  | 24.2  | 24.9  | 25.9  | 26.6 |      |
|                                                                               |                                                                               | MCDB   | 29.9  | 30.6  | 30.3  | 29.6  | 27.9  | 27.0  | 26.5  | 26.2  | 27.3  | 27.9  | 28.7  | 29.5 |      |
| 5%                                                                            |                                                                               | WB     | 26.5  | 27.5  | 27.4  | 26.5  | 25.2  | 24.0  | 23.2  | 22.9  | 23.5  | 24.2  | 25.2  | 26.1 |      |
|                                                                               |                                                                               | MCDB   | 29.3  | 30.1  | 29.8  | 29.0  | 27.3  | 26.2  | 25.7  | 25.5  | 26.6  | 27.5  | 28.1  | 29.2 |      |
| 10%                                                                           |                                                                               | WB     | 26.0  | 26.9  | 26.8  | 26.0  | 24.6  | 23.4  | 22.5  | 22.1  | 22.9  | 23.6  | 24.7  | 25.7 |      |
|                                                                               |                                                                               | MCDB   | 28.8  | 29.6  | 29.3  | 28.4  | 26.7  | 25.7  | 24.8  | 24.7  | 25.7  | 26.9  | 27.6  | 28.8 |      |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB                                                                         | MDBR   | 6.6   | 6.6   | 6.6   | 6.3   | 6.4   | 6.4   | 6.8   | 7.0   | 7.5   | 7.9   | 7.2   | 6.8  |      |
|                                                                               |                                                                               | MCDBR  | 7.6   | 7.4   | 7.6   | 7.4   | 6.9   | 7.0   | 7.7   | 7.8   | 8.4   | 8.4   | 7.7   | 7.5  |      |
|                                                                               |                                                                               | MCWBR  | 4.0   | 4.2   | 4.6   | 4.3   | 4.2   | 4.0   | 4.7   | 4.4   | 4.8   | 4.9   | 4.2   | 4.1  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 6.8   | 7.0   | 7.1   | 6.8   | 6.3   | 6.0   | 7.0   | 6.8   | 7.5   | 7.8   | 6.9   | 6.8  |      |
|                                                                               |                                                                               | MCWBR  | 4.3   | 4.8   | 4.8   | 4.5   | 4.4   | 4.0   | 4.8   | 4.5   | 5.2   | 5.2   | 4.8   | 4.3  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.401  | 0.393 | 0.390 | 0.384 | 0.390 | 0.375 | 0.355 | 0.363 | 0.372 | 0.384 | 0.387 | 0.385 |      |      |
|                                                                               | taud                                                                          | 2.553  | 2.581 | 2.601 | 2.608 | 2.546 | 2.586 | 2.621 | 2.590 | 2.546 | 2.534 | 2.551 | 2.583 |      |      |
|                                                                               | Ebn at Noon                                                                   | 945    | 941   | 920   | 887   | 842   | 839   | 870   | 895   | 923   | 938   | 951   | 960   |      |      |
|                                                                               | Edh at Noon                                                                   | 110    | 105   | 100   | 92    | 92    | 85    | 84    | 93    | 103   | 109   | 109   | 107   |      |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 6.58   | 6.37  | 5.56  | 4.67  | 3.95  | 3.57  | 3.81  | 4.49  | 5.48  | 6.09  | 6.43  | 6.77  |      |      |
|                                                                               | RadStd                                                                        | 0.47   | 0.43  | 0.35  | 0.40  | 0.32  | 0.30  | 0.24  | 0.25  | 0.48  | 0.43  | 0.45  | 0.44  |      |      |

## Historical Trends

|                            | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                            |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                            |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only          | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page